# Colter Snyder

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## SUMMARY

Software engineer with two years of professional back-end development in Python, Rust, and C++ on Linux, building containerized calculation services for scientific simulation platforms and the tooling to measure their performance. Comfortable from service code down to the hardware: designs FPGA logic in Verilog and writes 6502/Z80 assembly. Has taught data structures and algorithms in C++ to 120 students. M.S. in Computer Science (3.97 GPA).

## EXPERIENCE

**Innovative Scientific Solutions, Inc.**

*Software Engineer II* — *Dec 2024 – Present*

- Build Rust and Python (Flask) back-end calculation services with WebSocket messaging and MongoDB, deployed in Docker containers on Linux, for probabilistic nuclear weapons effects simulation platforms.
- Spearheaded an internal Python performance-benchmarking tool (a CLI, then a GUI front end) that runs any combination of the platform's 13 effect calculations and charts the results; it exposed a 10x slowdown in our interface to an external library, made the case to leadership with clear graphs, and saved an estimated several weeks of manual testing with hand-crafted JSON inputs.
- Port legacy physics models from Fortran and Python to Rust, preserving their numerical behavior while modernizing the codebase.
- Lead development of individual features on a 4–5 person agile team across the full lifecycle: refine requirements, write object-oriented designs, UML, and design documentation, implement, and verify with automated tests; take part in technical meetings with government customers.
- Review teammates' merge requests under a required separate-reviewer-and-merger process, and maintain code and system health with Git, GitLab CI/CD, SonarQube, and Python test tooling in ITAR/CUI environments.

**Tyler Technologies** — Lakewood, CO

*Software Development Intern* — *May 2023 – Aug 2023*

- Diagnosed and repaired a 2,000-test automated suite, restoring about 1,500 tests (75%) to passing and giving the team a reliable foundation for test-driven development.
- Migrated a major legacy VB6 desktop application to object-oriented C# as part of a cloud modernization effort.

**Colorado School of Mines** — Golden, CO

*Postgraduate Research Assistant, Department of Computer Science* — *May 2024 – Dec 2024*

- Developed object detection models for semiconductor cleanroom equipment and integrated them, with Python back-end services, into a Unity AR training app, delivering a working demo in collaboration with the University of Arizona (advisor: Dr. Mehmet Belviranli).

*Undergraduate Research Assistant* — *Oct 2019 – May 2020*

- Built a Python test suite as one of three developers on *Quality-aware Audio Multicast in Low Power Wireless Networks* (advisor: Dr. Qi Han).

## TEACHING

**Colorado School of Mines** — Golden, CO

*Adjunct Professor, Department of Computer Science* — *Jan 2023 – Dec 2023*

- Co-taught CSCI 220: Data Structures and Algorithms (C++) to 120 undergraduates, coaching students on algorithm design, complexity analysis, and debugging (Fall 2023).
- Taught a section of CSCI 102: Intro to Computer Science Lab (Python) to 90 undergraduates (Spring 2023).

## SKILLS

**Languages:** Python, C++, Rust, C, C#, Fortran, TypeScript, SQL, Verilog, 6502 & Z80 Assembly

**Systems & Tools:** Linux, Docker, Git, GitLab CI/CD, SonarQube, Flask, WebSockets, MongoDB

**Practices:** Data Structures & Algorithms, Performance Benchmarking & Visualization, Object-Oriented Design, Test-Driven Development, CI/CD, Agile

**Hardware Projects:** FPGA design (Verilog), 6502 & Z80 assembly programming, CAD/3D printing

## EDUCATION

**Colorado School of Mines** — Golden, CO

*Master of Engineering in Nuclear Engineering (in progress)* — *Aug 2026 – Expected May 2031*

*Master of Science in Computer Science • GPA 3.97* — *Jan 2022 – May 2024*

*Bachelor of Science in Computer Science • Magna Cum Laude • GPA 3.77* — *Aug 2019 – Dec 2022*

**Relevant coursework:** Theory of Computation, Cyber-Physical Systems Security, IoT Security & Privacy, Information Security & Privacy

**Honors:** CACI C-MAPP Scholar, National Merit Scholar, Dean's List, Eagle Scout, Rotary Youth Leadership Award